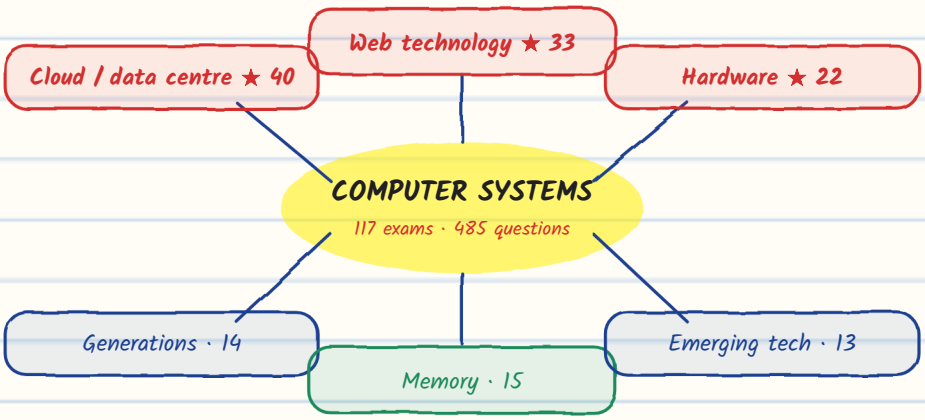
Introduction to Computer Systems

every computer-fundamentals, cloud, web and hardware question asked in BCS · NTRCA · BPSC · bank · power-sector · BUET-set papers, solved in 32 sheets



↑ the number is how many different exams asked that topic (red = most asked)

➜ How to read a question card

Q The question, as printed in the paper (typos fixed).
✎ **Asked in:** the exams it was taken from, e.g. BPSC AP 2022 · 45th BCS ICT (2024)

Every HTML / CSS / JS snippet and every Excel formula was run; green = the real output. Numerical answers (VM count, reliability) are worked on the sheet.

➜ Syllabus coverage

"Computer generations & classification, hardware & peripherals, memory, software, BIOS/booting, cloud computing & data centres, virtualization, web technology (HTML/CSS/JS, HTTP), e-commerce, emerging technology and ICT in society."

Most asked

① System vs application software · 4
② Cloud computing & service models · many
③ VM / hypervisor · 2+ (BB 2023)
④ Data-centre energy efficiency · 2
⑤ HTML element / page structure · CSS box model

Legend yellow = key term · red circle = must-memorise · **purple box** = memory trick · **red box** = exam trap · **green** = checked answer

➜ Contents

✓ Software · generations · types ... p. 2–4
✓ Building blocks · CPU · bus ... p. 5–6
✓ I/O devices · printers ... p. 7–8
✓ Memory · storage · data units p. 9–10
✓ BIOS & booting ..... p. 11
✓ Cloud: models · virtualization p. 12–13
✓ Data centre · scaling · security p. 14–16
✓ HTML I–II ..... p. 17–18
✓ CSS · JavaScript ..... p. 19–20
✓ Web / HTTP · hosting ... p. 21–22
✓ Emerging tech · e-commerce p. 23–24
✓ ICT society · banking IT ... p. 25–26
✓ Excel · graphics MCQs ... p. 27–28
✓ Rapid-fire · night-before · practice p. 29–32

1 System software vs application software

★ 4 exams — the single most-asked ICS question

Q1 What is the difference between system software and application software? / Explain software with examples. / Difference between hardware and software.

✎ **Asked in:** 45th BCS General (Sci & Tech) (2024) · BB AME 2016 · MM AP 2017 · NPCBL JAM 2022 · 40th BCS (2020) · 15th NTRCA Computer Operation (2019)

Point	System software	Application software
job	runs & manages the computer itself (always, in background)	does a user task (when opened)
examples	**OS** (Windows, Linux), compiler, driver, utility	**MS Word**, **Excel**, browser, banking app

Software = instructions that tell the hardware what to do (no physical form); **hardware** = the physical parts. A utility (antivirus, backup) is system software; **firmware** = software permanently in ROM (BIOS).

Q2 What is platform-independent software? Discuss with an example.

✎ **Asked in:** Combined Bank SO(IT) 2023

Platform-independent software runs on any OS/hardware unchanged, because it runs on a virtual machine, not the bare CPU. **Java**: source → `bytecode` → the **JVM** on each platform ("write once, run anywhere"). Also Python and any web page.

Q3 What is version control (e.g. Git)? Explain the difference between committing and pushing code.

✎ **Asked in:** Bank (name cut off) AP

Version control (Git) tracks code changes and lets teams roll back. **commit** = save a snapshot to your *local* repo; **push** = send local commits to the *remote* (GitHub). Commit is local, push is network.

Q4 Describe the classification of computer software. / List banking software used in Bangladesh and the features of good banking software.

✎ **Asked in:** Intl Crimes Tribunal Hardware Technician 2024 · Combined Bank Officer(IT) 2020

Software → **system** (OS, translators, utilities) + **application** (Office; custom core-banking, e.g. Flora, Ababil, T24). Good banking software: security, 24×7 uptime, real-time, audit trail, scalable, compliant.

Trap: a **compiler/assembler is system software** (serves the machine), not an app; **antivirus = utility = system software**.

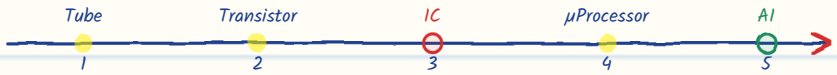
2 Generations of computers (1st–5th) 14 exams

Q5 What is a computer generation? Describe the characteristics of the different generations. / Give the features of the 2nd, 3rd and 5th generations.

✎ **Asked in:** 18th NTRCA School (2024) · 15th NTRCA Computer Operation (2019) · 13th NTRCA CSE (2016) · 17th NTRCA School (2023)

Gen	Switching device	Era	Features / example
1st	Vacuum tube	1946–59	huge, hot, machine language · ENIAC, UNIVAC
2nd	Transistor	1959–65	smaller, less heat, assembly/COBOL · IBM 1401
3rd	IC	1965–71	multiprogramming, keyboard/monitor · IBM 360
4th	Microprocessor (VLSI)	1971–now	PC, GUI, network · Intel 4004, IBM PC
5th	AI / ULSI	now→	parallel, AI, natural language · quantum

Trick: Tube → Transistor → IC → Microprocessor → AI. Each generation = smaller, faster, cheaper, less power.



↑ the switching device defines the generation

MCQ as asked	answer	asked in
Microprocessor was first used in which generation?	**Fourth (4th)**	DESCO AE(CSE) 2016
In which year were chips (ICs) first used inside a computer? 1964 / 1974 / 1975 / 1981	**1964** (3rd gen, IBM 360)	Combined Bank SO(IT) 2021
A webcam is a ___ device (System / Analogue / **Input** / Output)	**Input**	BB AD(General) 2023
The word "computer" means a ___ machine (হিসাবকারী / **গণনাকারী** / পরীক্ষার / বিমান)	**গণনাকারী (calculating)**	18th NTRCA College Prelim (2024)

Memory trick "Electricity" marker: vacuum tube (1), transistor (2), IC (3), microprocessor (4), AI (5). Any question naming a **transistor** ⇒ 2nd; **IC** ⇒ 3rd; **microprocessor** ⇒ 4th.

3 Analog / digital / hybrid · micro to super

Frequently tested (NTRCA)

Q6 Classify computers by working principle. / Difference between analog and digital computer. / In 3 fields, give the use of hybrid computers.

✎ **Asked in:** 17th NTRCA School (2023) · 15th NTRCA CSE

	Analog	Digital	Hybrid
works on	continuous physical quantity	discrete 0/1 digits	both
example	speedometer, thermometer	PC, laptop	ICU monitor, petrol pump
accuracy	lower	high	high

Hybrid use (3 fields): hospital ICU (measures BP/ECG as analog, shows digital), industrial process control, petrol pump, defence/aircraft.

Q7 Classify computers by size, shape and capacity. / Discuss super computer and mainframe. / Components of a micro-computer system.

✎ **Asked in:** 16th NTRCA CSE (2019) · 17th NTRCA School (2023) · BPSC ADA 2022

Type	Size / power	Use
Super	fastest, parallel	weather, nuclear, research (BD: BMD)
Mainframe	large, many users	bank, census, airline booking
Mini	medium	mid-size firm server
Micro (PC)	single user	desktop, laptop, mobile

Micro-computer system = microprocessor (CPU) + memory (RAM/ROM) + I/O devices + system bus + power supply, in one box. **Super vs mainframe:** super = max speed on one big job; mainframe = max throughput for many users/transactions.

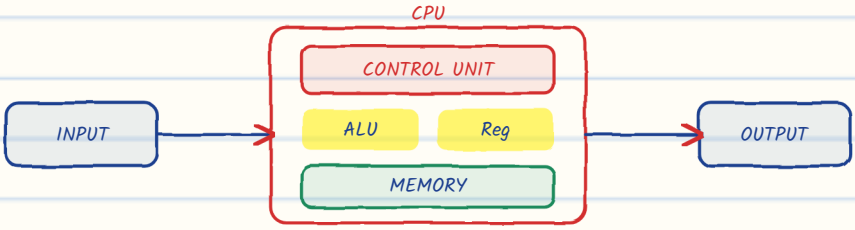
By purpose: general-purpose (PC) vs special-purpose (ATM, washing-machine controller — embedded). **Workstation** = a powerful single-user PC for engineering/graphics. **Server** = a computer that provides resources/services to clients over a network. **Embedded system** = a computer built into a device (car, router, smart TV).

Memory trick Order by power: **Micro < Mini < Mainframe < Super**. By signal: **Analog (continuous) · Digital (0/1) · Hybrid (both)**.

Exam trap A **mainframe** serves thousands of users at once; a **supercomputer** throws all its power at one scientific problem. They are not the same "big computer".

4 Building blocks & how a computer works BCS + NTRCA

Q8 What are the building blocks of a digital computer? Describe each unit briefly. / Draw the block diagram and describe each unit. / Explain the steps of a computer's working.
✎ **Asked in:** 38th BCS (2018) · 47th BCS CSE · 17th NTRCA CSE (2023) · 15th NTRCA CSE



↑ Input → CPU (CU + ALU + registers) ↔ Memory → Output; secondary storage holds data permanently

Units: Input (keyboard, mouse) feeds data; CPU = **Control Unit** (fetch-decode-control) + **ALU** (arithmetic + logic) + **registers**; Memory holds the program/data being used; Output (monitor, printer) shows results; **secondary storage** keeps data permanently. **Working = IPO(S):** Input → Process → Output → Storage; the CPU runs the **fetch-decode-execute** cycle.

Q9 What are the components of a micro-computer system? Explain the main organisational parts with a diagram.
✎ **Asked in:** BPSC ADA 2022 · 17th NTRCA CSE (2023) · 15th NTRCA CSE

CPU (microprocessor) + primary memory (RAM, ROM) + I/O devices + storage + the three **buses** (data, address, control) on the motherboard — see the bus sheet.

Q10 What is the total number of bits in IPv6? What do DIMM, TDP, SSD and HTTP stand for? / What criteria does computer speed depend on?
✎ **Asked in:** BUET ICT Cell AHE 2025 · BTCL JAM 2024

IPv6 = 128 bits. DIMM Dual In-line Memory Module · **TDP** Thermal Design Power · **SSD** Solid State Drive · **HTTP** HyperText Transfer Protocol. Computer speed depends on: clock speed (GHz), word size/bus width, number of cores, cache, RAM size and storage type (SSD vs HDD).

Fetch-decode-execute cycle: (1) the Control Unit **fetches** the next instruction from memory (address in the PC); (2) **decodes** it in the IR; (3) the ALU **executes** it; (4) the result is written back and the PC advances. This loop, millions/s, is all a computer does.

Memory trick Remember **IPO-S**: Input → Process (CPU) → Output → Storage. Every "how a computer works" answer is these four units plus the fetch-decode-execute cycle.

5 CPU, GPU, system bus, interrupts

hardware core

Q11 What is CPU? Describe the organisation of the CPU. / What is the function of a GPU?
Asked in: 17th NTRCA School (2023) · BPSC AP 2022

CPU ("brain") = **CU** (directs all operations) + **ALU** (arithmetic + logic/compare) + **registers** (fastest tiny stores: PC, IR, AC, MAR, MBR). GPU = many small cores for parallel graphics/matrix maths; used for 3D, video and **AI/ML** training. CPU = few fast general cores; GPU = thousands of simple cores.

Q12 What is a computer bus? What is the unit of its speed? Describe the types of bus. / Show how RAM, ROM and I/O connect to a microprocessor by bus. / Components of a USB bus with block diagram.
Asked in: BKKB AP 2017 · BPSC SAE 2021 · ICT Ministry AP 2017

Bus	carries	direction
Data bus	the actual data	two-way
Address bus	memory address	one-way (CPU→)
Control bus	read/write, clock, interrupt signals	two-way

A **bus** is a set of parallel wires carrying bits between CPU, memory and I/O. Speed is measured in MHz/GHz (bus clock) or bytes/s (bandwidth = width × clock). **USB** (Universal Serial Bus) = serial, hot-pluggable, powers the device; a USB packet = sync + PID + address/endpoint + data + CRC.

Q12b Write down the difference between a serial port and a parallel port.
Asked in: DESCO SAE 2023

Serial port sends 1 bit at a time over one wire (RS-232, USB) — fewer wires, good for long distance; **parallel port** sends many bits at once over many wires (old LPT/printer) — faster over short runs but bulky and prone to skew. Modern systems are nearly all serial (USB, SATA).

Q13 How can an interrupt be enabled? What steps occur when an interrupt is processed?
Asked in: 38th BCS (2018)

1. Device raises an **interrupt request (IRQ)** on the control bus (enabled by setting the interrupt-enable flag).
2. CPU finishes the current instruction, saves **PC + registers** on the stack.
3. Jumps to the **ISR** (interrupt service routine) via the interrupt vector.
4. ISR services the device, then **IRET** restores the saved state and resumes.

Q14 Differentiate among Intel Core i3, i5 and i7 processors.
Asked in: Combined Bank SO(IT) 2023

Rising cores/threads, cache and Turbo Boost: **i3** entry (2–4 cores, no/low turbo), **i5** mid (4–6 cores, turbo), **i7** high (6–8+ cores, hyper-threading, big cache). More cores + cache + turbo = faster but costlier/hotter.

6 Optical & magnetic input: OCR, OMR, MICR, RFID

bank favourite

Q15 What is OCR? Difference between OCR and OMR. / Distinguish between OMR and MICR. / Name two optical input devices.

✎ **Asked in:** NPCBL JAM 2022 · BB AME 2016 · BPSC ADA 2022

Device	Full form / reads	Use
OCR	Optical Character Recognition — printed/handwritten **characters**	scan text to editable text
OMR	Optical Mark Reader — pencil **marks**	MCQ/OMR answer sheets
MICR	Magnetic Ink Character Recognition — magnetic ink digits	bank cheques
Barcode/RFID	light / radio waves	POS, inventory, toll

MICR is at the bottom of a cheque (E-13B font); it survives stamps/signatures because it is read magnetically, not optically — that is why banks use it. **OMR** reads *where* a mark is (not what it says); **OCR** reads the characters.

Q16 RFID has huge applications in supply-chain and toll collection. Show the basic working principle of RFID.

✎ **Asked in:** Combined Bank Officer(IT) 2020

RFID = Radio-Frequency IDentification: a **tag** (chip + antenna) + a **reader**. The reader sends radio waves; a passive tag draws power from them and transmits its stored ID back; the reader sends it to the system. No line-of-sight needed (unlike barcode). Use: toll (e-toll), library, inventory, access cards.

MCQ as asked	answer	asked in
Which device is used for primary input of OCR?	**Scanner**	BB AD(General) 2018
In MICR, "C" stands for —	**Character**	Teletalk Executive 2019
Which is NOT contained in a MICR code line?	**Signature** (it has bank/branch/account/cheque no.)	Combined Bank Officer(IT) 2020
A bar-code reader emits —	**Light (laser/LED)**	Combined Bank Officer(IT) 2020
Which is true for RFID?	**No line-of-sight; uses radio waves**	Combined Bank Officer(IT) 2020

7 Printers, monitor, resolution

hardware MCQ + written

Q17 Explain impact vs non-impact printer. / Describe the working of an inkjet / laser printer with a diagram. / What is a non-impact printer?

Asked in: Combined Bank SO(AME) 2021 · Combined Bank AHE 2021 · 18th NTRCA School-2 (2024) · 15th NTRCA Computer Operation (2019)

	Impact	Non-impact
prints by	hammer/pins strike an inked ribbon	ink/toner, no contact
examples	dot-matrix, line, daisy-wheel	**inkjet**, **laser**, thermal
noise / speed	noisy, slow	quiet, fast
carbon copy?	**yes** (multi-part)	no

Inkjet: tiny nozzles spray charged ink droplets onto paper. **Laser:** a laser draws the image on a charged drum → drum picks up toner → toner transferred to paper → fused by heat. **Dot-matrix (impact)** printers are used at bank cash counters because they make carbon copies of vouchers.

Q18 What do you understand by the resolution of a computer screen? / Factors to consider when buying a monitor. / How does an optical mouse work?

Asked in: BPSC AP 2022 · BB AME 2016 · 36th BCS (2016)

Resolution = number of pixels (width × height, e.g. 1920×1080); more pixels = sharper. Buy a monitor by: size, resolution, panel (IPS/VA), refresh rate, response time, ports, eye comfort. **Optical mouse:** an LED lights the surface, a tiny camera takes thousands of images/s, a DSP computes the movement.

MCQ as asked	answer	asked in
Which printer is used at a bank cash counter / in banks & offices?	**Dot-matrix** (counter) · **Laser** (office)	Combined Bank Officer(IT) 2026
Printing quality is measured in — (ppm / **dpi** / cpi / cps)	**dpi** (dots per inch)	BB AD(General) 2023
Which first when troubleshooting a faulty monitor?	**Check its connections** to PC & power	Combined Bank AHE/AE(IT) 2021
___ helps prevent power surges	**Surge protector / UPS**	Combined Bank SO(IT) 2021

Exam trap "ppm" is printer speed (pages per minute); **dpi** is print quality. A camera for bank surveillance is a **CCTV / IP camera**.

8 RAM / ROM, SSD / HDD, optical disk 15 exams

Q19 Difference between RAM and ROM. / Why is RAM unsuitable as permanent storage?
Asked in: 15th NTRCA CSE · BTRC AD(Technical) 2019 · BTRC SAD 2021

	RAM	ROM
volatile?	yes — lost on power off	no — permanent
read/write	read & write	read only (mostly)
holds	running programs/data	BIOS/firmware
types	SRAM (cache), DRAM (main)	PROM, EPROM, EEPROM

RAM is unsuitable as permanent storage because it is volatile — it loses its contents when power is off. Memory hierarchy (fast→slow, small→big): **registers → cache → RAM → SSD/HDD**.

Q20 Classify common storage devices. What is a solid-state device? / How does an optical disk read/write data? / What is MIDI? Describe the structure of a CD-ROM.
Asked in: 45th BCS ICT (2024) · BPSC ANE 2020 · 15th NTRCA Computer Operation (2019)

	HDD (magnetic)	SSD (solid-state)	Optical (CD/DVD)
stores by	magnetising platters	charge in flash cells	pits burned, read by **laser**
moving parts	yes (arm, platter)	none	spins
speed / shock	slower, fragile	fast, durable	slow

SSD = flash memory, no moving parts → faster, silent, durable. **Optical** disk: a laser burns/reads pits & lands on a spiral track; CD → DVD → Blu-ray = shorter wavelength, more capacity. **MIDI** = Musical Instrument Digital Interface (stores notes/commands, not audio).

MCQ as asked	answer	asked in
Which is main memory? RAM / ROM / HDD / Floppy	**RAM**	DoICT AP 2020
SSDs are more durable in harsh environments because —	**They have no actuator arms / moving parts**	Combined Bank SO(IT) 2021
EDSAC stored data using what memory?	**Mercury delay lines**	Combined Bank ADA 2021
Best way to protect hard-drive data	**Regular backup**	Combined Bank AHE/AE(IT) 2021

9 Bits, bytes, units; data vs information

MCQ staple

Q21 Difference between data and information. / Difference between MB and Mbps. / Discuss data-processing methods.

✎ **Asked in:** SPCBL SAE 2022 · W2PDCL SAE 2023 · 17th NTRCA School (2023)

Data = raw, unorganised facts; **information** = processed, meaningful data. **MB** (megabyte) = a storage size (8 bits = 1 byte); **Mbps** (megabits per second) = a speed. 1 byte = 8 bits, so 100 MB ≠ 100 Mb. Data processing: collection → input → process → output → storage (batch, online, real-time, distributed).

Unit	=	Unit	=
1 byte	8 bits	1 GB	1024 MB
1 KB	1024 B	1 TB	1024 GB
1 MB	1024 KB	nibble	4 bits

MCQ as asked	answer	asked in
A Unicode character uses how many bits? 8/16/20/24	**16**	BCC AP 2021
1 TB equals how many bytes?	**10^{12} (SI)**; 2^{40} = 1,099,511,627,776 (binary)	BB AD(General) 2022
Exact bytes in 960 Kb main memory	**960 × 1024** bytes	BPSC AP 2022
1 megabyte = how many kilobytes? 1000/512/1026/1024	**1024**	BPSC AME 2017
Correct order by size	**byte < KB < MB < GB < TB**	BDCCL AM(Transmission) 2022
Unit of hard-disk capacity (MHz/kHz/**Gigabyte**/None)	**Gigabyte**	BCC AP 2021
Typical speed of USB 3.0	**4.8 Gbit/s** (≈ 5 Gbps)	Combined Bank SO(IT) 2021
GPS stands for	**Global Positioning System**	BCC AP 2021

Number systems (covered in the Number Systems lesson): binary (base 2), octal (8), decimal (10), hex (16). Character codes: **ASCII** = 7/8-bit (English); **Unicode (UTF-8/16)** covers all scripts incl. Bangla; **EBCDIC** is the old IBM 8-bit code. Full units chain: bit → nibble (4) → byte (8) → KB → MB → GB → TB → PB → EB.

Exam trap In computing, 1 KB = **1024** bytes (binary), but disk makers use SI 1 KB = 1000 B — which is why a "1 TB" drive shows ~931 GB. Write both forms if the options allow.

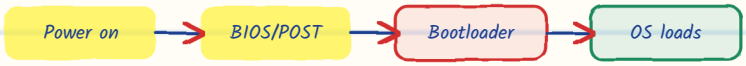
10 BIOS, booting, POST, CMOS

6 exams

Q22 What is BIOS? What is its main role in a computer? / Explain BIOS in a server and how it affects booting in hardware maintenance.

✎ **Asked in:** EGCB SAE(ICT) 2023 · 45th BCS General (Sci & Tech) (2024) · RAKUB ANSE 2023

BIOS = Basic Input/Output System — **firmware stored in ROM** on the motherboard. On power-on it: (1) runs **POST** (Power-On Self-Test) to check RAM, keyboard, disk; (2) initialises hardware; (3) finds the boot device and loads the **bootloader** → the OS. Settings (boot order, date, passwords) are stored in **CMOS**, kept alive by a small battery. On servers, BIOS/UEFI sets RAID, boot order and virtualization flags — wrong settings stop the OS loading.



↑ the boot sequence: a single beep = POST passed

MCQ as asked	answer	asked in
BIOS stands for —	**Basic Input / Output System**	BTCL JAM 2022
Which is a WRONG statement about BIOS? (connects CPU & I/O / manages data flow / loads OS / **provides storage**)	"**Provides storage**" (it does not)	Combined Bank AHE/AE(IT) 2021
No beep on boot — what to check first?	**Power supply**	Combined Bank AHE/AE(IT) 2021
How to clear a CMOS password?	**Remove the BIOS battery / change the jumper**	Combined Bank AHE/AE(IT) 2021

BIOS vs UEFI: UEFI is the modern replacement — supports disks > 2 TB (GPT), a graphical setup, faster boot and Secure Boot. **Cold boot** = starting from power-off; **warm boot** = restarting without cutting power (Ctrl+Alt+Del). **Bootstrapping** = the small program that "pulls the system up by its bootstraps" to load the OS.

Memory trick Order on power-on: **Power** → **POST** → **BIOS** → **Bootloader** → **OS**. One short beep = all hardware OK; repeated beeps = a hardware fault (RAM/GPU) by beep code.

11 Cloud computing & IaaS / PaaS / SaaS

★ the biggest ICS topic — 40 exams

Q23 What is cloud computing? Write its advantages and disadvantages / service models. / How does it differ from traditional computing? / Write the three basic functions of cloud services.

✎ **Asked in:** 17th NTRCA CSE (2023) · 17th NTRCA ICT (2023) · NPCBL JAM 2022 · BPSC AP 2021 · Parliament Secretariat SAE 2023 · NPCBL Engineer(Cyber Security) 2026 · BOF AE(CSE) 2021 · BLPU AP 2018

Three basic functions / essential characteristics of cloud services (NIST): on-demand self-service, broad network access and resource pooling (plus rapid elasticity and measured/pay-per-use service).

Cloud computing = delivering computing resources (servers, storage, databases, software) **over the internet, on demand, pay-as-you-go**, instead of owning hardware. **Advantages:** no upfront cost, elastic scaling, anywhere access, automatic backup/updates, high availability. **Disadvantages:** needs internet, data-security/privacy concerns, vendor lock-in, recurring cost, less control. **Deployment:** public, private, hybrid, community.

Model	You get	You manage	Example
IaaS	VMs, storage, network	OS, runtime, app	AWS EC2, Azure VM
PaaS	+ OS & runtime platform	just your app/code	Heroku, App Engine
SaaS	ready software	nothing (just use)	Gmail, Office 365

Trap scenario (Combined Bank SO(IT) 2026): a startup wants to deploy code only, not manage OS/runtime/hardware → PaaS (e.g. Heroku, Google App Engine). **5 enabling technologies of cloud:** virtualization, broadband network, distributed/grid computing, service-oriented architecture, web services/APIs, automation. Pricing/deploy often: **IaaS** = rent infrastructure; **SaaS** = rent software.

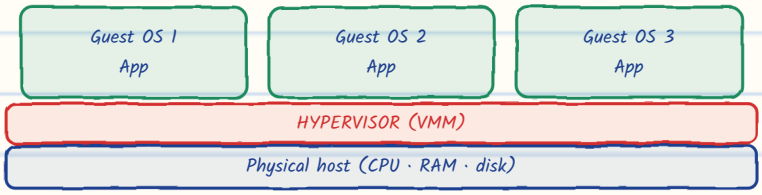
MCQ as asked	answer	asked in
Which is NOT a layer of cloud computing? (CaaS / IaaS / PaaS / SaaS)	**Computing as a service (CaaS)**	Combined Bank AHE/AE(IT) 2021
Which will NOT be treated as a cloud service?	**A web portal self-hosted by PSC**	BPSC AME 2022
Not a reason to use colocation for DR instead of building a DC	**Full control of hardware**	Combined Bank ADA 2021

12 Virtual machine, hypervisor, container

★ BB AME + AP 2023, BUET AHE 2025

Q24 Define a virtual machine with a diagram, explain how a VM works, and list its benefits. / What is a hypervisor and its types? / Define virtualization; role of hypervisor; VM vs container.

✎ **Asked in:** BB AME 2023 · BB AP 2023 · BUET ICT Cell AHE 2025 · Combined Bank AP(SB & Janata) 2026 · RAKUB ANSE 2023 · Parliament Secretariat SAE 2023



↑ one physical host runs several isolated VMs, each with its own guest OS, over a hypervisor

VM = a software computer with its own virtual CPU/RAM/disk and **its own guest OS**, running on a **hypervisor (VMM)** that shares the real hardware. **Benefits:** isolation, run many OSes on one box, easy snapshot/backup/recovery, fast provisioning, better utilisation. **Hypervisor types: Type 1 (bare-metal)** runs directly on hardware (ESXi, Hyper-V, Xen) — fast, for servers; **Type 2 (hosted)** runs on a host OS (VirtualBox, VMware Workstation) — for desktops.

	Virtual machine	Container (Docker)
includes	a full guest OS	app + libs only, shares host OS kernel
size / start	GBs, boots in minutes	MBs, starts in seconds
isolation	strong (own OS)	lighter (process-level)

Docker = container platform; **Docker Hub** = cloud image repository. **Kubernetes (K8s)** orchestrates containers (control plane + worker nodes/pods). **JVM** runs Java bytecode portably — it is not a hypervisor.

Q25b For reliable, efficient data transmission, VMs or containers? / Which architecture for a system monitoring a third-party cloud service through APIs?

✎ **Asked in:** BSCL AE(CSE) 2026

VMs vs containers: prefer **containers** — lower overhead, faster start, easy scaling and CI/CD; use VMs only for strong OS-level isolation or a different guest OS. **API monitoring:** an **event-driven microservices** design (services subscribing to the APIs + a queue + a dashboard) scales and isolates faults.

Q25 A physical server has 32 CPU cores, 96 GB RAM, 4 TB storage. Each VM needs 4 cores, 16 GB RAM, 500 GB. Max VMs without over-committing? Which resource is the limit?

✎ **Asked in:** Combined Bank Officer(IT) 2026

CPU: 32/4 = **8** · RAM: 96/16 = **6** · Storage: 4000/500 = **8**. Max = min(8, 6, 8) = **6 VMs**; **RAM is the limiting resource.**

13 Data centre, energy, maintenance, DR

bank AME/AHE core

Q26 What are the challenges in optimising the energy efficiency of a data centre? / Explain the components of a data centre. / Key hardware components of a typical server.
Asked in: BB AME 2023 · BB AP 2023 · ABL SO(IT) 2017 · Rupali Bank ANA 2023

Data-centre components: servers (compute), storage/SAN, network (switches, routers, firewall), power (UPS, generator, PDU), cooling (CRAC/HVAC), racks, fire suppression, physical security. **Energy-efficiency challenges:** cooling is the biggest draw, low average server utilisation, power-distribution losses, legacy/inefficient equipment, measuring and managing **PUE** (Power Usage Effectiveness = total power ÷ IT power; ideal → 1.0). Fixes: virtualization/consolidation, hot/cold-aisle containment, free cooling, efficient PSUs.

Q27 Distinguish preventive vs corrective maintenance in a data centre (with an example). / What should be checked before buying servers / a data centre? / DR & business-continuity plan for your data centre.
Asked in: Combined Bank AME 2026 · Combined Bank ADA 2026 · Combined Bank ADA 2024 · Combined Bank AME/AHE 2023 · Combined Bank SO(IT) 2024

	Preventive	Corrective
when	before failure, scheduled	after a failure
example	cleaning, patching, replacing aging UPS battery	swap a dead disk, restart a crashed service
goal	avoid downtime	restore service fast

Before buying servers/a DC: workload/capacity, CPU/RAM/storage, redundancy (power, cooling, network), scalability, Tier level, warranty/support, security, cost/TCO, location. **DR/BCP:** define **RTO** (how fast to recover) & **RPO** (how much data loss is acceptable), backups (3-2-1 rule), a hot/warm/cold standby site, regular drills. **Reliability** of 3 series components each $0.95 = 0.95^3 =$ 0.857

Q28 What is the need for an edge server? / What do you mean by a National Data Center?
Asked in: Petrobangla AM(CSE) 2024 · BPSC AP 2022

Edge server processes data near the source (not a far cloud): lower latency, less bandwidth, real-time response, local privacy (CDN, IoT, CCTV). **National Data Center** (BD: Kaliakoir, Tier-4) hosts government data/services centrally with high security and uptime.

14 Scaling, web-app deployment, big data

BUET AHE + bank

Q29 Difference between vertical and horizontal scaling. What is auto-scaling and how does it work? / Draw a server rack showing horizontal and vertical scaling.

✎ **Asked in:** BUET ICT Cell AHE 2025 · RPGCL AM(ICT) 2022

	Vertical (scale up)	Horizontal (scale out)
means	bigger CPU/RAM in the same server	more servers/nodes behind a load balancer
limit	one machine's max	near-unlimited
downtime	usually needs restart	none (add nodes live)

Auto-scaling watches load (CPU/requests) and automatically adds nodes when busy and removes them when idle, so you pay only for what you use and stay available at peak.

Q30 Describe a typical web-application deployment architecture and how the tiers interact. / Sketch a plan to transform a data centre into a cloud. / What is an API Gateway and how does it interact with backend microservices?

✎ **Asked in:** Combined Bank AME/AHE 2023 · Combined Bank AME/AE(IT) 2024 · Combined Bank ADA 2026

3-tier web app: client (browser) → **web/presentation tier** (load balancer + web servers) → **application tier** (business logic, APIs) → **data tier** (database). An API Gateway is the single entry point in front of microservices: it routes requests, authenticates, rate-limits, aggregates responses and hides the internal services. **DC→cloud plan:** assess apps → choose model (IaaS/PaaS) → pick strategy (rehost/replatform/refactor) → migrate data → test → cut over → optimise.

Q31 Write ten sentences on "Big Data". What is Big Data and its characteristics? / Identify 5 Hadoop-ecosystem components for processing massive server logs. / SaaS & multi-tenant architecture: advantages and disadvantages.

✎ **Asked in:** MRA SCO 2023 · BPSC AP 2021 · Combined Bank AME 2026 · Combined Bank SO(IT) 2024

Big Data = data sets too large/fast/varied for ordinary tools; the **5 V's**: Volume, Velocity, Variety, Veracity, Value. **Hadoop ecosystem: HDFS** (storage), **MapReduce/YARN** (processing/resource mgmt), **Hive** (SQL on big data), **HBase** (NoSQL), **Spark** (fast in-memory), **Kafka** (log/stream ingestion). **Multi-tenant SaaS** = one app instance serves many customers (tenants); pro = low cost, easy updates; con = weaker isolation, "noisy neighbour", harder per-tenant customisation.

15 Cloud security, threats, server scenarios

scenario essays

Q32 How do security assessments and audit reports detect vulnerabilities and ensure cloud-security-posture compliance? / Cyber-security threats to a microfinance app hosted in your DC, and how to mitigate them. / Tools/technology to secure an e-banking web server in your DC.
✎ **Asked in:** BSCL AE(CSE) 2025 · Combined Bank SO(IT) 2024 · Combined Bank AME/AE(IT) 2024

Cloud-security posture: regular vulnerability scans + penetration tests + audit logs + config review (CSPM) find mis-configurations, open ports, weak IAM, unpatched software, and check compliance (ISO 27001, PCI-DSS). **Threats** to a hosted app: malware/ransomware, SQL injection & **XSS**, DDoS, data breach, insider misuse, phishing. **Mitigation:** firewall + WAF, HTTPS/TLS, strong IAM & MFA, patching, input validation, encryption at rest/in transit, backups, IDS/IPS, least privilege, monitoring.

Q33 Server maintenance best practices (cleaning, monitoring, patching) and how they help longevity. / How do you handle server-related problems for your company? / Explain BIOS in a server and its effect on boot config.
✎ **Asked in:** Rupali Bank ANA 2023 · Rupali Bank ANE 2023 · Combined Bank Officer(IT) 2023 · RAKUB ANSE 2023

Server best practices: keep firmware/OS patched, monitor (CPU/RAM/disk/temperature, alerts), clean dust & check cooling, test backups & failover, review logs & security, document changes, redundant power/RAID. These extend hardware life and keep uptime high.

Q33b A company's server traffic rises from 10,000 to 50,000 hits but it does not move to the cloud. Discuss the business and technical consequences.
✎ **Asked in:** ICB Securities AP 2026

Technical: on a fixed on-prem server, 5× load causes CPU/RAM saturation, slow response, timeouts, downtime at peak, no elasticity, and costly over-provisioning for peaks. **Business:** poor user experience, lost transactions/revenue, reputation damage, higher capital & maintenance cost, and slow time-to-scale vs a cloud auto-scaling competitor.

Scenario / MCQ as asked	answer	asked in
HTTP status code 500 indicates —	**Internal Server Error**	BB AD(ICT) 2025
Server upload rejected (unsupported file format): which HTTP code?	**415 Unsupported Media Type** (or 400)	Combined Bank ADA 2026
Cloud outage: cloud-provider vs bank responsibility	**Shared responsibility**: provider = infra uptime/SLA; bank = data, access, its own DR	Combined Bank ADA 2026
What is Docker & Docker Hub?	**Docker = container platform; Hub = cloud image repo**	BB AD(ICT) 2025

16 HTML elements, page structure, HTML5 tags

★ 17th NTRCA ×2 + banks

Q34 What is an HTML element? Explain with examples. / Explain the structure of an HTML web page with an example.

✎ **Asked in:** 17th NTRCA CSE (2023) · 17th NTRCA ICT (2023)

An **HTML element** = a start tag + content + end tag, e.g. `<p>Hello</p>`. Some are empty (`<br>`, `<img>`). A page's skeleton:

```
<!DOCTYPE html>
<html>
 <head><title>My Page</title></head>
 <body>
  <h1>My First Heading</h1>
  <p>My first paragraph.</p>
 </body>
</html>
```

Renders: a big bold "My First Heading" then the line "My first paragraph." (a W2PDCL SAE 2023 trace: a red paragraph via `style="color:red"` shows in red).

Q35 Describe the HTML5 tags <header> <footer> <section> <article>. / Name HTML5 media tags. / How do you change an HTML element's value using the DOM?

✎ **Asked in:** Combined Bank AME 2019 · CB AP 2018

Tag	meaning	Tag	meaning
<header>	top/intro, logo, nav	<section>	a thematic group
<footer>	bottom, copyright	<article>	self-contained content

HTML5 media tags: `<audio>`, `<video>`, `<source>`, `<track>`, `<embed>`. **Change a value via DOM:** `document.getElementById("demo").innerHTML = "New";` (`localStorage` keeps data with no expiry; `sessionStorage` clears when the tab closes — ABL Officer(ICT) 2017). **<canvas>** draws pixels via JS (good for games); **SVG** is vector XML (scales, each shape is in the DOM).

Short Q / MCQ as asked	answer	asked in
Which tag creates a button in HTML?	**<button>** (or <input type="button">)	BCC AP 2023
Which tag is used to write JavaScript in HTML?	**<script>**	BCC AP 2023
The newest version of HTML is —	**HTML5**	Combined Bank Officer(IT) 2020

17 Tables, lists, forms (GET/POST), web structure

NTRCA written staple

Q36 Write HTML for a page with a hyperlink, an ordered list, an unordered list and a form (text box, check box, option, button). / Write the HTML for the given nested list.

✎ **Asked in:** 15th NTRCA ICT · 17th NTRCA ICT (2023) · 17th NTRCA CSE (2023)

```
<a href="page.html">Go</a>
<ol><li>Fruits<ol type="a"><li>Mango</li><li>Orange</li></ol></li></ol>
<ul><li>Vegetables<ul><li>Green capsicum</li></ul></li></ul>
<form method="post">
 <input type="text"> <input type="checkbox">
 <input type="radio" name="g"> <button>Submit</button>
</form>
```

Renders: a link, a numbered list (1. Fruits → a. Mango, b. Orange), a bullet list, and a form row with a text field, a checkbox, a radio button and a Submit button.

Q37 Write HTML for the given table (with rowspan/colspan). / Make a table structure. / What is a website? Describe linear-sequence and tree web structures.

✎ **Asked in:** 18th NTRCA College ICT (2024) · 16th NTRCA ICT · DUET AP 2020

```
<table border="1">
 <tr><th>Name</th><th>Product</th></tr>
 <tr><td rowspan="3">Mr. John</td><td>HP</td></tr>
 <tr><td>ASUS</td></tr><tr><td>Dell</td></tr>
</table>
```

rowspan/colspan merge cells down/across. **Web structures:** *linear* = pages in a fixed sequence (slideshow, sign-up steps); *tree/hierarchical* = a home page branching to sections (most sites).

Q38 Difference between GET and POST; when to use each (with example). / What is web hosting / a web portal? Why is hosting needed?

✎ **Asked in:** 15th NTRCA ICT · BAERA Engineer(CSE) January 2021 · 16th NTRCA ICT

	GET	POST
data in	URL (query string, visible)	request body (hidden)
size / cache	limited, cacheable, bookmarkable	large, not cached
use for	search/read (idempotent)	login, submit, upload

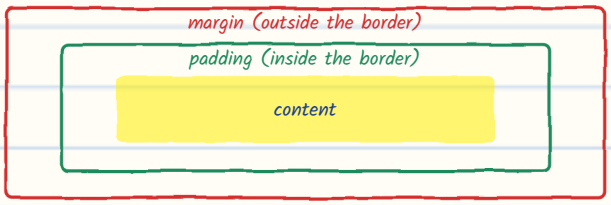
Web hosting = renting server space to store a site so it is reachable 24×7 on the internet; needed because a site must live on an always-on, public server. A **web portal** is a gateway page aggregating many services (e.g. a government portal).

18 CSS: box model, types, frameworks, responsive

★ CSS box model repeats

Q39 In the CSS box model, difference between the Padding and Margin areas. / What is CSS? What is a CSS framework? Name three. / What are the types of CSS?

✎ **Asked in:** 15th NTRCA ICT · 17th NTRCA ICT (2023) · BEPZA Programmer 2023 · 18th NTRCA College ICT (2024)



↑ content → padding → border → margin

Padding = space *inside* the border, between content and border (part of the element's background). Margin = space *outside* the border, separating this element from others (transparent). **CSS** (Cascading Style Sheets) styles HTML. **Three types:** inline (`style=""`), internal (`<style>` in `<head>`), external (`.css` file — best). **Frameworks:** Bootstrap, Tailwind, Foundation, Bulma.

Q40 Explain the role of CSS media queries and the viewport meta tag in responsive apps. What is "mobile-first"? / Why use CSS in a website?

✎ **Asked in:** Combined Bank AME/AE(IT) 2026(Jan) · NESCO AM(ICT) 2021 · ICT Ministry AP 2021

```
<meta name="viewport" content="width=device-width, initial-scale=1">
@media (max-width: 600px) { .menu { flex-direction: column; } }
```

The **viewport meta tag** makes the page use the device's real width (no zoomed-out desktop view on phones). **Media queries** apply different CSS at different screen widths. **Mobile-first** = design the small-screen layout first, then add rules for larger screens with `min-width` queries. **Why CSS:** separates style from content, consistent look, one change updates every page, faster loading, responsive design.

Exam trap Padding counts as part of the clickable/background box; margin does not. With `box-sizing: border-box`, width includes padding + border (not margin).

19 JavaScript, PHP, web-input security

banks + power sector

Q41 Write JS with if-else so that for x = 1..9 it prints "x is an odd value." / "x is an even value." / Difference between == and === in JavaScript.

✎ **Asked in:** PGCB SAE 2021 · BAERA Engineer(CSE) January 2021

```
for (let x = 1; x <= 9; x++) {
  if (x % 2 === 0) console.log(x + " is an even value.");
  else console.log(x + " is an odd value.");
}
```

1 is an odd value. · 2 is an even value. · … · 9 is an odd value. (odd for 1,3,5,7,9; even for 2,4,6,8)

== compares values with type coercion (5 == "5" → true); === compares value and type, no coercion (5 === "5" → false). Always prefer ===.

Q42 What is a closure in JavaScript? Explain with an example. / Write a JS function to validate a 10-digit phone number (digits only, no spaces/+).

✎ **Asked in:** ICB AP 2017 · NESCO AM(ICT) 2021

```
function phone(s){ return /^\d{10}$/.test(s.replace(/\D/g,"")); }
console.log(phone("01712345678")); // 11 digits -> false
console.log(phone("1712345678"));  // 10 digits -> true
```

false, true. A **closure** = an inner function that remembers variables from its outer function even after the outer returns (used for private state, counters).

Q43 A CMS blog body inserts un-escaped HTML as {{body}}, so a <script> can be injected. Explain the risk and how to prevent it (XSS), with CSP enabled.

✎ **Asked in:** DPDC AM(ICT) 2025

Risk = stored XSS (Cross-Site Scripting): if user content like {{body}} is placed into the page without escaping, a malicious post can run attacker JavaScript in every visitor's browser (steal cookies/sessions, deface, act as the user). **Prevention** (the real exam answer): (1) output-encode / sanitise all user content — escape < > & " ' or use a sanitiser (DOMPurify); never inject raw HTML; (2) a strict Content-Security-Policy that blocks inline and third-party scripts; (3) set cookies **HttpOnly** + Secure + SameSite so script cannot read them; (4) validate input and use framework auto-escaping. (We describe the defence, not a working attack payload.)

Short Q as asked	answer	asked in
What is PHP? Why is JavaScript needed in web development?	**PHP** = server-side scripting (dynamic pages, DB); **JS** = client-side interactivity/validation	ICT Ministry AP 2021

20 HTTP status codes, SOAP/REST, protocols

bank written

Q44 What do the HTTP status-code categories 2xx, 4xx, 5xx represent? Give an example of each. / What does status code 500 indicate?

Asked in: Combined Bank AME/AE(IT) 2026(Jan) · Combined Bank AME 2026 · BB AD(ICT) 2025

Class	meaning	example
1xx	informational	100 Continue
2xx	success	200 OK, 201 Created
3xx	redirect	301 Moved, 304 Not Modified
4xx	client error	404 Not Found, 401, **415**
5xx	server error	**500 Internal Server Error**

Q45 What are SOAP and RESTful APIs? State one main difference in how they exchange data. / What is SOAP? / Three differences between POST and GET.

Asked in: Combined Bank Officer(IT) 2025 · BB AP 2019 · BAERA Engineer(CSE) January 2021

	SOAP	REST
data format	XML only, strict	JSON/XML/text, flexible
style	protocol, has WSDL	architectural style over HTTP
weight	heavy	**lightweight, fast**

Main difference: **SOAP always exchanges XML messages** (with an envelope); **REST** uses plain HTTP verbs (GET/POST/PUT/DELETE) and usually lightweight JSON.

Q46 Browsing www.psc.gov.bd — list the protocols and their layers used to fetch the page, and the address types at Transport, Network and Data-Link layers.

Asked in: Parliament Secretariat SAE 2023

DNS (application) resolves the name → IP; **HTTP/HTTPS** (application) requests the page; **TCP** (transport, uses port numbers); **IP** (network, uses IP addresses); **Ethernet/Wi-Fi** (data-link, uses MAC addresses). HTTPS adds TLS for encryption.

MCQ as asked	answer	asked in
Which is NOT a web server? (Apache Tomcat / **PHP** / Jetty)	**PHP** (a language, not a server)	BB AD(ICT) 2025
URL stands for —	**Uniform Resource Locator**	BB AD(General) 2018

21 Going live, deployment, static vs dynamic

mixed web

Q47 Once a website is designed & developed, what are the steps to make it live online? / Create a webpage and deploy it to an Apache server. / Difference between static and dynamic websites.

✎ **Asked in:** Teletalk Executive 2019 · BUET ICT Cell AHE 2025 · 17th NTRCA ICT (2023)

1. Buy a **domain name** and a **hosting** plan (or a cloud VM).
2. Build the site; put files in the web root (Apache: `/var/www/html`).
3. Point the domain's **DNS** (A record) to the server IP.
4. Configure the web server (Apache/Nginx), enable **HTTPS** (SSL cert).
5. Upload via FTP/Git, test, then it is live; monitor & back up.

	Static site	Dynamic site
content	fixed HTML, same for all	generated per request (DB, user)
tech	HTML/CSS/JS	+ server (PHP, Node) + DB
example	a brochure page	Facebook, a bank portal

Q48 True/false on HTTP: one page with text + 3 images needs how many request/response messages? Can two sites share one persistent connection? Can one TCP segment carry two HTTP messages? Do responses ever have an empty body?

✎ **Asked in:** Combined Bank SO 2020

(i) **True** — 1 request + 1 response for the HTML, then 1 request + 1 response per image = 4 requests, 4 responses. (ii) **False** — a persistent connection is to one host; different hosts need different connections. (iii) **False** — HTTP messages are not split that way across one segment by design. (iv) **False** — responses like 204/304 have an empty body.

Short Q as asked	answer	asked in
What makes it easier to find a visited web page later?	**History / Bookmarks**	BB AD(General) 2018
The first page of a website (in the address bar)	**Home page**	BB AD(General) 2018
Importance of an application framework in web development	**Reusable structure**: faster, secure, maintainable code	BB AME 2019

Q48b Create 3 tables in an Oracle DB; build one web page with 3 buttons — "Show students" (list from a table), "Show subject-wise marks" (join subject + marks), "Show each student's total" (join all three tables).

✎ **Asked in:** BUET AP 2025

Pattern: three tables (`student`, `subject`, `marks`); a page with 3 buttons, each calling a server endpoint that runs one SQL query and renders the result table. Button 2 **joins** subject + marks on `subject_id`; button 3 joins all three and does `GROUP BY student` with `SUM(mark)` for each student's total. (The SQL join detail is drilled in the DBMS set.)

22 Blockchain, nanotech, IR 4.0, quantum

13 exams

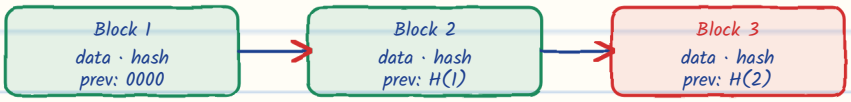
Q49 Write about blockchain: how it works, benefits, uses. / What is nanotechnology, its uses? / What is IR 4.0 and its key components?

✎ **Asked in:** W2PGCL AE(CSE) 2023 · Combined Bank SO(IT) 2023 · 17th NTRCA CSE (2023) · ICT Ministry AP 2021 · Fisheries & Livestock Instructor(CS) 2023

Blockchain = a distributed, append-only ledger of blocks, each holding data + a timestamp + the previous block's **hash**, copied across many nodes. Changing one block breaks every later hash, so it is **tamper-evident**; consensus (PoW/PoS) adds new blocks. **Benefits:** decentralised, transparent, immutable, secure, no middleman. **Uses:** cryptocurrency, smart contracts, supply chain, land records, e-voting.

Nanotechnology = engineering matter at 1–100 nm (atomic/molecular scale). Uses: medicine (targeted drug delivery), electronics (smaller chips), materials (stronger/lighter), energy, water filtration. **IR 4.0** (Fourth Industrial Revolution) fuses the physical, digital and biological worlds; **key components:** AI, IoT, Big Data, cloud, robotics/automation, cyber-physical systems, 3D printing, AR/VR. Unlike IR 1–3 (steam, electricity, computers), IR 4.0's change is **exponential and interconnected**. **Quantum computer** uses qubits (superposition + entanglement) → massively parallel, vs a normal computer's definite 0/1 bits.

Emerging tech	one-line definition
IoT	everyday devices with sensors connected to the internet, sharing data
AI / ML	machines that learn from data to make decisions
AR / VR	overlay digital on real (AR) / fully simulated world (VR)
3D printing	additive manufacturing, layer by layer from a digital model



↑ each block stores the previous block's hash — change one block and every later hash breaks

Memory trick Each block = **data + own hash + previous hash**. The "previous hash" chain is what makes a blockchain tamper-evident — remember the chain, not the coins.

23 E-commerce, WWW vs internet, search engines

NTRCA + BCS

Q50 What is e-commerce? Its types? / Classify e-commerce by nature of transaction. / Difference between e-commerce and m-commerce. / Barriers to e-commerce.

✎ **Asked in:** NESCO AM(ICT) 2018 · BPSC ANE 2020 · 17th NTRCA CSE (2023) · 37th BCS · Combined Bank AHE 2021

Type	meaning / example	Type	meaning / example
B2C	business→consumer (Daraz)	C2C	consumer→consumer (Bikroy)
B2B	business→business (Alibaba)	G2C	govt→citizen (e-gov)

E-commerce = buying/selling goods & services online. **M-commerce** = e-commerce on a mobile device (apps, mobile wallet, location-based). **Barriers:** trust/security, low card/internet penetration, logistics/delivery, legal framework, digital literacy, payment-gateway limits.

Q51 What is the WWW and how does it differ from the internet? / E-commerce vs e-business; e-governance.

✎ **Asked in:** Parliament Secretariat SAE 2023 · 15th NTRCA CSE

The internet is the global network of networks (the hardware + TCP/IP). The WWW is one service running on it — interlinked web pages accessed by HTTP via browsers. Internet also carries email, FTP, etc. **E-business** is wider than e-commerce (all online business processes, not just buying/selling). **E-governance** = delivering government services electronically (G2C, G2B, G2G).

MCQ as asked	answer	asked in
Bangladesh's first internet search engine	**পিপীলিকা (Pipilika)**	16th NTRCA College Prelim (2019)
Which is the first search engine?	**Archie** (1990)	Combined Bank Officer(IT) 2020
Which is NOT a search engine?	**Facebook** (it is social media)	BB AD(General) 2021

Search engine parts: a crawler/spider fetches pages, an indexer builds a searchable index, and a query/ranking engine returns results (Google uses PageRank). Examples: Google, Bing, Yahoo, DuckDuckGo; BD: Pipilika (first Bangla search engine, by SUST, 2013). First-ever: Archie (1990).

Exam trap Facebook, YouTube and Gmail are **not search engines**. "Browser" (Chrome) ≠ "search engine" (Google): the browser is the app, the search engine is the service you open in it.

24 E-governance, Smart Bangladesh, MIS

descriptive, Secretariat/ICT

Q52 Write short notes: e-Governance; Smart Bangladesh. / Challenges to building Digital Bangladesh. / Challenges to applying MIS in Bangladesh. / Modern-life applications of the internet.

✎ **Asked in:** Parliament Secretariat SAE 2023 · Intl Crimes Tribunal Hardware Technician 2024 · 17th NTRCA School (2023)

E-Governance = using ICT to deliver government services and improve transparency/efficiency; four types: **G2C** (citizen), **G2B** (business), **G2G** (govt-govt), **G2E** (employee). Smart Bangladesh (Vision 2041) builds on Digital Bangladesh with four pillars: Smart Citizen, Smart Economy, Smart Government, Smart Society. **Digital Bangladesh challenges:** infrastructure & electricity, affordability, digital literacy, cyber-security, local content, the digital divide.

MIS (Management Information System) turns data into reports for managers' decisions. **MIS challenges in BD:** poor data quality, lack of skilled staff, resistance to change, weak infrastructure, high cost, no integration between systems, data-security gaps. **Internet in modern life:** education, e-commerce, banking (MFS), health (telemedicine), government services, communication, remote work/freelancing, entertainment.

Key BD ICT landmarks: a2i (Access to Information), e-GP (electronic government procurement), Union Digital Centres, the National Data Center (Kaliakoir, Tier-4), the ICT Act 2006 / Digital Security Act, and the "Digital Bangladesh" (2008) → "Smart Bangladesh" (Vision 2041) programme. **Fourth Industrial Revolution vs IR 1–3:** IR1 steam, IR2 electricity/mass production, IR3 computers/automation, IR4 fuses physical–digital–biological systems (AI, IoT, cyber-physical) — change is exponential and networked.

e-Governance type	between	BD example
G2C	govt → citizen	e-passport, land records, MyGov
G2B	govt → business	e-GP (e-tender), VAT online
G2G	govt → govt	inter-ministry file (e-Nothi)
G2E	govt → employee	iBAS++ payroll

Model-answer frame For any "short note / describe" ICT-society question: **definition → 3–4 features/pillars → BD examples (a2i, MFS, e-GP, Union Digital Centre) → one line on challenges.**

25 Digital banking, electronic payment systems

bank written favourite

Q53 Difference between digital and traditional banking. How can digital banking promote financial inclusion? / Describe the transformative power of ICT for online banking (ten points).
✎ **Asked in:** Combined Bank SO(IT) 2023 · Combined Bank AME/AHE 2023 · Combined Bank AME/AE(IT) 2024

	Traditional banking	Digital banking
channel	branch, in person	app / web, 24×7
reach	limited by branches	anywhere with a phone
cost / speed	higher, slower	lower, instant

Financial inclusion: MFS (bKash, Nagad), agent banking and mobile accounts bring unbanked/rural people into the system — low cost, no branch needed, instant remittance, small savings/loans, government disbursements (allowances, subsidies) direct to a phone.

Q54 Define an electronic payment system (EPS) with a diagram; name five types. / Difference between traditional and electronic payment systems; types of EPS.
✎ **Asked in:** BPSC AP 2023 · Intl Crimes Tribunal Hardware Technician 2024

EPS = paying for goods/services electronically, no cash. **Flow:** customer → merchant → **payment gateway** → acquiring bank → card network → issuing bank → back. **Five types:** card (debit/credit), **mobile financial services** (bKash/Nagad), internet/online banking, e-wallet, e-cash/crypto; also NFC/QR. **Traditional vs electronic:** cash/cheque, physical, slow vs digital, instant, trackable, needs internet & security.

Term	meaning (bank IT)
Payment gateway	software that authorises & routes a card/online payment between merchant and banks
NPSB / RTGS / BEFTN	BD inter-bank rails: card switch · real-time large-value · batch EFT
MFS	Mobile Financial Services (bKash, Nagad, Rocket)
KYC	Know Your Customer — identity verification before an account

Model-answer frame EPS essay: definition → the gateway flow (6 actors) → 5 types with a BD example each → security (encryption, OTP, 2FA, PCI-DSS) → one line on fraud risk.

26 Excel: formulas, VLOOKUP, IF, errors

NESCO + banks

QSS Cells A1:A4 hold four plants' power output. Write a function in A5 for the average of the best three. / Explain the VLOOKUP formula. / Using IF, put the smallest of A1,B1,C1 into D1.

✎ **Asked in:** NESCO AM(ICT) 2018 · BTCL JAM 2024 · PGCB SAE 2021

> A5: =AVERAGE(LARGE(A1:A4,{1,2,3}))
> D1: =MIN(A1,B1,C1) (or nested IF)
> VLOOKUP: =VLOOKUP(lookup_value, table_range, col_index, FALSE)

=AVERAGE(LARGE(A1:A4,{1,2,3})) returns the mean of the three largest of the four cells (the exact exam answer). VLOOKUP finds a value in the first column of a range and returns a value from the given column number; FALSE = exact match.

MCQ as asked	answer	asked in
In Excel, every formula starts with —	**= (equals sign)**	Combined Bank Officer(IT) 2026
=Min(Average(B1,B2), Product(B3:B4)), B1=20 B2=24 B3=15 B4=20	Average=22, Product=300 → **Min = 22**	BB AD(General) 2023
#NAME? error occurs when —	**Function/name is misspelled** (ক); #REF! = bad reference	BB AD(General) 2023
User-developed instructions for Excel are called —	**Macro**	BB AD(General) 2021
Center-align text shortcut in MS Word	**Ctrl + E**	BB AD(General) 2023

Common Excel functions: SUM, AVERAGE, COUNT, COUNTIF, MAX/MIN, LARGE/SMALL, IF, AND/OR, VLOOKUP/HLOOKUP, INDEX+MATCH, CONCATENATE, LEFT/RIGHT/MID, ROUND, NOW/TODAY. **Reference types:** relative A1, absolute A1 (fixed when copied), mixed $A1. **Errors:** #NAME? bad name/function · #REF! deleted reference · #DIV/0! divide by zero · #VALUE! wrong type · #### column too narrow.

VLOOKUP pitfalls: the lookup column must be the first column of the range; use FALSE for an exact match (TRUE needs a sorted table); col_index counts from the range's first column, not the sheet's. For a left-side lookup use INDEX+MATCH. **Office shortcuts:** Ctrl+C/V copy/paste · Ctrl+Z undo · Ctrl+E center (Word) · F12 Save As · Ctrl+S save.

Exam trap =Product(B3:B4) multiplies (15×20=300); Average(B1,B2) = 22; Min picks the smaller → 22, not 300. Read the nesting from the inside out.

27 Transformations: reflection, rotation, shear

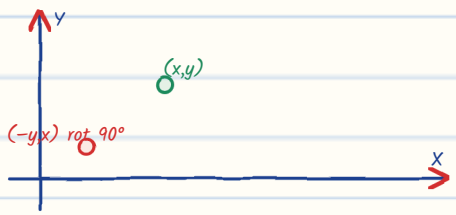
BUET M.Sc. admission MCQs

These appeared in the BUET M.Sc. admission test (counted as a job exam). 2D/3D transformation rules:

Transform	rule (2D/3D)
Reflection on XZ plane	(x, y, z) → (x, **−y**, z)
Rotation 90° about Z	(x, y) → **(−y, x)**
Shear by k on X axis	x' = x + k·y, y' = y

MCQ as asked	answer	asked in
A(3,4,1) reflected on the XZ plane →	**(3, −4, 1)**	BUET M.Sc. April 2021
Point (1,2,3): rotate 90° about Z axis →	**(−2, 1, 3)**	BUET M.Sc. April 2021
Triangle (1,1),(0,0),(1,0), shear k=2 on X →	**(3,1),(0,0),(1,0)**	BUET M.Sc. April 2021
Plane equidistant from P1(0,0,0) & P2(2,2,0)	**x + y − 2 = 0**	BUET M.Sc. April 2021
Orthographic projection: face is ___ to the picture plane	**Parallel**	BUET M.Sc. April 2021

Worked — rotate (1,2) 90° about Z: (x,y)→(−y,x) = (−2,1); z unchanged → **(−2,1,3)** **Shear check:** x'=x+2y → (1+2·1, 1)=(3,1); (0,0)→(0,0); (1,0)→(1,0).



↑ rotate 90° about Z: (x, y) → (−y, x); reflection on XZ flips y only

Exam trap Reflection on the **XZ plane** flips the **y** sign only (not x). Reflection on XY flips z; on YZ flips x. Match the missing axis to the flipped coordinate.

28 Keyboard, Office & device MCQs

bank general + BB AD

MCQ as asked	answer	asked in
Shift, Ctrl and Alt keys are called —	**Modifier keys**	BB AD(General) 2026
Which is a modifier key of the keyboard?	**Ctrl / Shift / Alt**	Combined Bank Officer(IT) 2020
Which operation does F1 perform in most apps?	**Help**	Combined Bank Officer(IT) 2020
What does F3 stand for in most computers?	**Search / Find next**	BB AD(General) 2021
Key to show the Save-As box in MS Word	**F12**	BB AD(General) 2021
Which is an extension of a video file format?	**.avi / .mov / .mp4**	BB AD(General) 2021
Which is NOT a video file format? AVI/MOV/**JPEG**/JPG	**JPEG / JPG** (image)	BPSC AP 2022
Which is used as an image-file extension? .docx/.xls/**.jpg**	**jpg**	BTCL JAM 2022
Which technology is used in a Compact Disk (CD)?	**Laser**	BPSC AP 2022
The job of ___ is to translate an array of dots into text	**OCR software**	Combined Bank Officer(IT) 2020
In a plasma display, gas is electrified by a grid of —	**Electrodes**	Combined Bank Officer(IT) 2020
In a spreadsheet, ___ helps you make sense of a worksheet's contents	**Chart / graph**	Combined Bank Officer(IT) 2020
OTG cable is NOT related to —	**Printing** (OTG links phone ↔ USB device)	BB AD(General) 2018
A device that is both input and output	**Touch screen / modem / network card**	BB AD(General) 2018
You can include a picture/object by using —	**Insert menu**	BB AD(General) 2018
Mixed tick-mark set: fastest memory? / which SQL runs last? / HTTP error port?	**Register (fastest) · ORDER BY/COMMIT last · port 80 (HTTP)**	BAERA Engineer(CSE) January 2021
Programs that help a computer manage its tasks/devices are called ___	**System software**	BPSC AME 2019
Which is a utility software? Windows / Oracle / **Antivirus** / MS Word	**Antivirus**	BPSC AME 2017
First Bangla software for the visually impaired (আইলিশ / আইসাইট / **আইডট** / আইলাইট)	**আইডট (i-dot)**	17th NTRCA School Prelim (2022)

Exam trap **JPEG/JPG, PNG, GIF, BMP** = image; **AVI, MOV, MP4, MKV, WMV** = video; **MP3, WAV** = audio; **DOCX, XLSX, PDF** = document. Sort the odd one out by media type.

29 Internet, storage & misc MCQs BCC · banks · BPSC

MCQ as asked	answer	asked in
Which is NOT a web browser? Firefox/**Facebook**/Chrome/Safari	**Facebook**	BPSC AP 2022 · BTCL JAM 2022
In which year did Gmail start? 1998/1988/**2004**/2021	**2004**	BCC AP 2021
Which is a correct email format?	**name@domain.com**	BCC AP 2021
Which language helps you learn Android programming?	**Java** (also Kotlin)	BCC AP 2021
Full form of DVD	**Digital Versatile Disc**	BB AD(General) 2018
When browsing, the browser stores some data as a —	**Cookie**	DESCO AE(CSE) 2016
Which is NOT a true statement about the Recycle Bin?	**"Disk space increases by sending files to it"** (false)	Combined Bank AHE/AE(IT) 2021
.NET can be used for —	**All of the above** (desktop, microservices/containers, cloud)	Combined Bank SO(IT) 2021
Java Virtual Machine —	**Manages memory & gives Java a portable execution environment**	Combined Bank SO(IT) 2021
Main task for a computer in mapping geographical data	**Data retrieving and drawing**	BPSC AP 2022
Why is RAM unsuitable as permanent storage?	**Volatility**	BTRC AD(Technical) 2019 · BTRC SAD 2021

Q56 Write short notes on DNS, cloud server and firewall. / Business Intelligence model. / Decision tree for playing football under weather conditions. / Ergonomics: how to judge a good chair. / SQL query to retrieve the uncommon names from two tables.

✎ **Asked in:** Intl Crimes Tribunal Hardware Technician 2024 · Combined Bank SO(IT) 2023 · BAERA Engineer(CSE) 2023 · Combined Bank SO(IT) 2020 · RAKUB ADBA 2020

DNS maps names→IP; **cloud server** = a virtual server rented over the internet; **firewall** filters traffic by rules. **BI** turns data into dashboards/reports for decisions (ETL → data warehouse → OLAP → visualisation). **Decision tree**: branch on weather (outlook/humidity/wind) to a Play/Don't-Play leaf. **Ergonomics** (good chair): adjustable height, lumbar support, armrests, breathable, feet flat, screen at eye level. **Uncommon names** (symmetric difference) = names in exactly one table (SQL: `FULL JOIN … WHERE a IS NULL OR b IS NULL`, or two `EXCEPT` queries `UNION`ed).

★ Night before the exam

read in 10 minutes

1. **System vs application software** (★ most asked): system runs the machine (OS, compiler, driver, utility); application does a user task (Word, browser). Antivirus = utility = system software.
2. **Generations**: tube → transistor → IC → **microprocessor (4th)** → AI. Each = smaller, faster, cheaper.
3. **CPU** = CU + ALU + registers; **buses** = data (2-way) / address (1-way) / control. IPv6 = 128 bits.
4. **OCR** characters · **OMR** marks · **MICR** magnetic cheque digits · **RFID** radio, no line-of-sight.
5. **RAM** volatile, read/write · **ROM** permanent, holds BIOS · **SSD** no moving parts, fast.
6. 1 byte = 8 bits; 1 KB = 1024 B (binary) / 1000 (SI). Unicode char = 16 bits. MB = storage, Mbps = speed.
7. **BIOS** firmware in ROM → POST → bootloader → OS; settings in CMOS (battery).
8. **Cloud**: on-demand resources over the internet. **IaaS** rent infra · **PaaS** deploy code only · **SaaS** ready app. Deploy-only startup ⇒ PaaS.
9. **VM** = own guest OS on a hypervisor (Type 1 bare-metal / Type 2 hosted); **container** shares the host kernel (Docker), lighter. VM count = min over CPU/RAM/storage.
10. **Data centre**: cooling is the biggest energy cost; PUE → 1 is ideal. Preventive (before) vs corrective (after) maintenance. DR = RTO + RPO + backups.
11. **Scaling**: vertical = bigger box; horizontal = more boxes + load balancer; auto-scaling adds/removes nodes.
12. **HTML** element = tags + content; HTML5 semantic: header/footer/section/article. **CSS** box: content→padding→border→margin; 3 types inline/internal/external.
13. **GET** in URL, cacheable, read · **POST** in body, for login/submit. **HTTP**: 2xx ok, 4xx client, 5xx server (500 = internal error).
14. **SOAP** = XML only; **REST** = lightweight over HTTP verbs. **XSS** fix: escape output, CSP, HttpOnly cookies.
15. **Blockchain** = block (data + hash + previous hash), immutable chain. **IR 4.0** = AI + IoT + Big Data + cloud + robotics.
16. **E-commerce**: B2C, B2B, C2C, G2C. **EPS** flow: customer → gateway → acquirer → network → issuer. **WWW** ⊂ internet.

✓ Practice MCQs (all asked; answers at the bottom)

exam pattern

1. Which is NOT a layer of cloud computing? (a) CaaS (b) IaaS (c) PaaS (d) SaaS ✎ Combined Bank AHE/AE(IT) 2021

2. BIOS stands for: (a) Basic Input/Output System (b) Basic Interrupt/Output System (c) Binary I/O Set (d) None ✎ BTCL JAM 2022

3. A Unicode character uses: (a) 8 bits (b) 16 bits (c) 20 bits (d) 24 bits ✎ BCC AP 2021

4. HTTP status code 500 means: (a) Bad Request (b) Unauthorized (c) Internal Server Error (d) Not Found ✎ BB AD(ICT) 2025

5. Microprocessor was first used in which generation? (a) 2nd (b) 3rd (c) 4th (d) 5th ✎ DESCO AE(CSE) 2016

6. Which is NOT a web browser? (a) Firefox (b) Facebook (c) Chrome (d) Safari ✎ BPSC AP 2022

7. Which tag writes JavaScript in HTML? (a) <js> (b) <script> (c) <style> (d) <code> ✎ BCC AP 2023

8. The CSS area outside the border is: (a) padding (b) content (c) margin (d) outline ✎ 15th NTRCA ICT

9. Which is main memory? (a) HDD (b) ROM (c) RAM (d) Floppy ✎ DoICT AP 2020

10. In Excel, every formula starts with: (a) + (b) = (c) @ (d) # ✎ Combined Bank Officer(IT) 2026

Answers: 1 a · 2 a · 3 b · 4 c · 5 c · 6 b · 7 b · 8 c · 9 c · 10 b

➜ ✎ Written practice (model answer marked)

Q What is cloud computing? Explain IaaS, PaaS and SaaS. (10)
✎ **Asked in:** Combined Bank AP(SB & Janata) 2026 · DMTCL AE(ICT) 2023 · BB AME 2023

Cloud computing = on-demand delivery of computing resources over the internet, pay-as-you-go (2).
IaaS = rent VMs/storage/network, you manage OS & app — AWS EC2 (2).
PaaS = platform + runtime, you deploy only your code — Heroku, App Engine (2).
SaaS = ready software, just use it — Gmail, Office 365 (2).
Add a diagram (who-manages-what stack) + benefits/one example each (2).

Memory trick ICS answers: **definition → table/diagram → BD example → one trap line.** A cloud/VM answer always needs the who-manages-what stack or the hypervisor diagram.

Before the exam ✓ service models (IaaS/PaaS/SaaS), VM vs container, HTTP 2xx/4xx/5xx, CSS box model, OCR/OMR/MICR, generations, RAM vs ROM, byte units.